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Quick Activity 1: School Supplies Calculator

1. Identify the inputs, processes, and outputs.

Inputs	Processes	Outputs
Number of notebooks	Calculate the notebook and pen costs.	Total cost
Number of pens	Add the costs to get the total cost.	Change
Number of pens	Subtract the total cost from the payment to get the change.	

2. Divide the problem into smaller tasks. Write the tasks in their correct order.

- Start the program.
- Enter the number of notebooks.
- Enter the number of pens.
- Enter the amount of payment.
- Calculate the cost of the notebooks.
- Calculate the cost of the pens.
- Add both costs to get the total cost.
- Subtract the total cost from the payment.
- Display the total cost and change.
- End the program.

3. Present your solution using either pseudocode or a flowchart.

START

```
READ notebooks, pens, payment
notebookAmount ← notebooks × 45
penAmount ← pens × 12
purchaseTotal ← notebookAmount + penAmount
moneyBack ← payment – purchaseTotal
DISPLAY purchaseTotal
DISPLAY moneyBack
```

END

4. Perform a dry run using both test cases below. Show each calculation and the resulting values.

Step	Variable	Calculation	Result
1	notebookAmount	$3 \times \text{₱}45$	₱135
2	penAmount	$2 \times \text{₱}12$	₱24
3	purchaseTotal	$\text{₱}135 + \text{₱}24$	₱159
4	moneyBack	$\text{₱}200 - \text{₱}159$	₱41

Total Cost: ₱159

Change: ₱41

Step	Variable	Calculation	Result
1	notebookAmount	$2 \times \text{₱}45$	₱90
2	penAmount	$3 \times \text{₱}12$	₱36
3	purchaseTotal	$\text{₱}90 + \text{₱}36$	₱126
4	moneyBack	$\text{₱}150 - \text{₱}126$	₱24

Total Cost: ₱126

Change: ₱24